Volume between Two Paraboloids

Stewart, *Calculus: Early Transcendentals*, Section 15.7, Exercise 20

Problem

Use a triple integral to find the volume of the given solid.

The solid enclosed by the paraboloids $y = x^2 + z^2$ and $y = 8 - x^2 - z^2$.

Part (a)

This exercise has a single part. The whole solution is Part (a), and the supporting files are named with `part1`.

The symbols. E is the solid region that the two paraboloids enclose. $\mathrm{d}V$ is the volume element, that is, the volume of a very small piece of E. The coordinates are x, y, and z, and the vertical axis in this exercise is y. The letter D denotes the projection of E onto the xz-plane, the horizontal plane. Inside that plane we use polar coordinates: r is the distance from the origin of the xz-plane, so $r^2 = x^2 + z^2$, and θ is the angle measured from the positive x-axis toward the positive z-axis. Then $x = r\cos\theta$ and $z = r\sin\theta$, and the area element of D is $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$. The symbol $\bar{y}$ below is the average value of y on E. This exercise uses no spherical coordinates (ρ, θ, φ), and the letter ρ appears only once below, as a shorthand for the distance from the y-axis, so there it means the same quantity as r.

Step 1. Describe the solid

Both surfaces are surfaces of revolution about the y-axis, because each one depends on x and z only through the combination $x^2 + z^2$. The surface $y = x^2 + z^2$ is a bowl that opens toward the positive y-axis and touches the origin. The surface $y = 8 - x^2 - z^2$ is a cap that opens toward the negative y-axis and reaches $y = 8$ on the y-axis.

The two surfaces meet where the two right-hand sides are equal:

$$x^2 + z^2 = 8 - x^2 - z^2 \iff 2(x^2 + z^2) = 8 \iff x^2 + z^2 = 4. \tag{1}$$

So the intersection is the circle of radius 2 in the horizontal plane $y = 4$. On both surfaces the height at that circle is $y = 4$.

Write $\rho^2 = x^2 + z^2$ for short, so ρ is the distance from the y-axis. Then the lower surface is $y = \rho^2$ and the upper surface is $y = 8 - \rho^2$, and their difference is

$$(8 - \rho^2) - \rho^2 = 8 - 2\rho^2 = 2(4 - \rho^2). \tag{2}$$

This difference is positive for $\rho < 2$ and is zero at $\rho = 2$. Therefore the upper surface lies above the lower surface exactly over the disk of radius 2, and E is one solid piece. The region is

$$E = \{(x, y, z) : x^2 + z^2 \le 4,\ x^2 + z^2 \le y \le 8 - x^2 - z^2\}. \tag{3}$$

The projection of E onto the horizontal xz-plane is the disk

$$D = \{(x, z) : x^2 + z^2 \le 4\}. \tag{4}$$

Figure 1 shows the two paraboloids, the intersection circle, and one vertical slice of the solid.

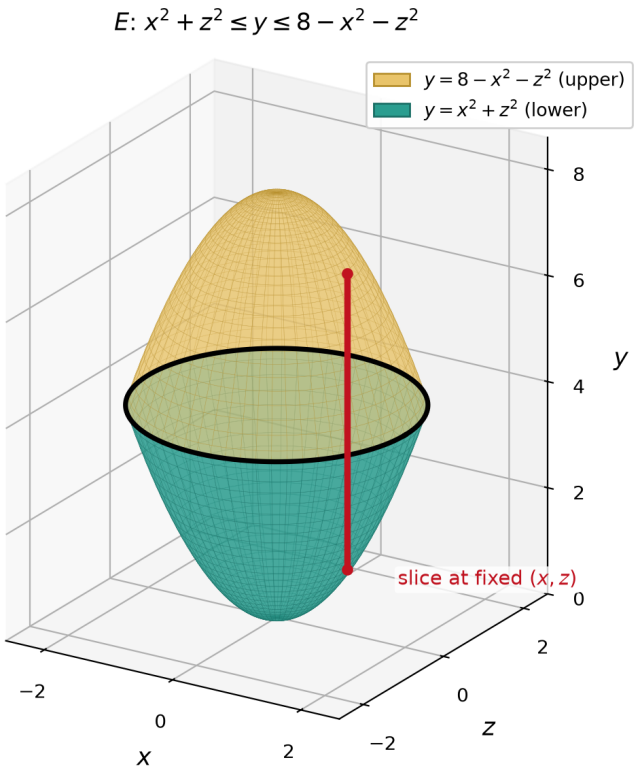


Figure 1: The solid E of Exercise 15.7.20. The gold surface is the upper cap $y = 8 - x^2 - z^2$, drawn for $x^2 + z^2 \le 4$. The teal surface is the lower bowl $y = x^2 + z^2$ over the same disk. The heavy black curve is the intersection circle $x^2 + z^2 = 4$ at height $y = 4$. The red segment is the slice of E at one fixed pair (x, z), and it runs from $y = x^2 + z^2$ up to $y = 8 - x^2 - z^2$. The vertical axis of the picture is y.

Step 2. Choose the order of integration

The volume of E is the triple integral of 1 over E:

$$V(E) = \iiint_E 1 \, dV. \tag{5}$$

In Equation (3) the variable y appears alone, and both of its bounds are functions of (x, z). A straight line parallel to the y-axis enters E at $y = x^2 + z^2$, leaves E at $y = 8 - x^2 - z^2$, and meets

E in exactly one segment for every interior point (x, z) of D. So E is simple with respect to y, and we integrate in y first:

$$V(E) = \iint_D \left(\int_{x^2+z^2}^{8-x^2-z^2} 1 \, \mathrm{d}y \right) \mathrm{d}A = \iint_D \big(8 - 2(x^2 + z^2)\big) \, \mathrm{d}A. \tag{6}$$

This order is legal by Fubini's theorem, because the integrand 1 is constant and the two bounds are continuous functions of (x, z) on D.

Step 3. Compute the inner integral

The inner integral has length equal to the height of the solid above the point (x, z):

$$\int_{x^2+z^2}^{8-x^2-z^2} 1 \, \mathrm{d}y = \Big[y \Big]_{y=x^2+z^2}^{y=8-x^2-z^2} = (8 - x^2 - z^2) - (x^2 + z^2) = 8 - 2(x^2 + z^2). \tag{7}$$

Equation (7) is already Equation (2). At the centre of the disk the height is 8. At the rim of the disk, where $x^2 + z^2 = 4$, the height is 0. The height decreases steadily with the distance from the y-axis.

Step 4. Convert the outer integral to polar coordinates

The region D of Equation (4) is a disk, and the integrand $8 - 2(x^2 + z^2)$ depends on x and z only through $x^2 + z^2$. Both facts call for polar coordinates in the xz-plane. With

$$x = r \cos\theta, \qquad z = r \sin\theta, \qquad \mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta, \tag{8}$$

the disk becomes the rectangle

$$0 \le r \le 2, \qquad 0 \le \theta \le 2\pi, \tag{9}$$

because $x^2 + z^2 \le 4$ gives $r \le 2$, and a full disk needs the whole turn $0 \le \theta \le 2\pi$. Figure 2 shows the cross section of E in the plane $z = 0$ and the disk D with one polar element.

The integrand becomes

$$8 - 2(x^2 + z^2) = 8 - 2r^2 = 2(4 - r^2). \tag{10}$$

The product of Equations (8) and (10) is the full element of the integral:

$$\big(8 - 2r^2\big) r \, \mathrm{d}r \, \mathrm{d}\theta = \big(8r - 2r^3\big) \, \mathrm{d}r \, \mathrm{d}\theta. \tag{11}$$

The extra factor r comes from the area element. Without that factor the integral would be wrong, so write it down before any antiderivative.

Step 5. Evaluate the integral

Equation (11) does not contain θ, so the integral in θ is 2π times the integral in r:

$$V(E) = \int_0^{2\pi} \int_0^2 \big(8r - 2r^3\big) \, \mathrm{d}r \, \mathrm{d}\theta \tag{12}$$

$$= 2\pi \int_0^2 \big(8r - 2r^3\big) \, \mathrm{d}r. \tag{13}$$

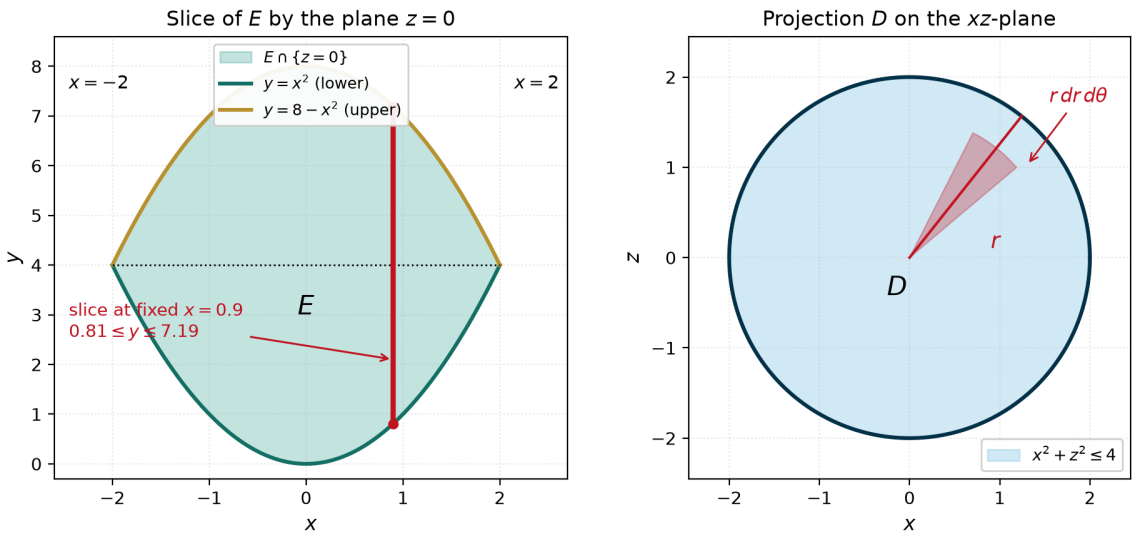


Figure 2: Two views of the solid of Exercise 15.7.20. Left: the cross section of E by the plane $z = 0$. The region lies between the parabola $y = x^2$ and the parabola $y = 8 - x^2$, for $-2 \le x \le 2$, and the two parabolas meet at $(\pm 2, 4)$. The red segment is the slice at $x = 0.9$. Right: the projection D of E onto the xz-plane is the disk of radius 2. The shaded sector is one element of area $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$.

The antiderivatives are $4r^2$ for $8r$ and $r^4/2$ for $2r^3$. Hence

$$\int_0^2 \left(8r - 2r^3\right) \mathrm{d}r = \left[4r^2 - \frac{r^4}{2}\right]_0^2 = (16 - 8) - 0 = 8. \tag{14}$$

The volume is therefore

$$V(E) = 2\pi \cdot 8 = 16\pi. \tag{15}$$

Answer

$$\boxed{V(E) = 16\pi \approx 50.2655}$$

Step 6. Check the value

The value was checked twice. The first check is exact and uses SymPy with a different order of integration. The second check is numerical and uses NumPy only, with four independent methods.

Exact check in the order x, then z. From Equation (6),

$$V(E) = \int_{-2}^{2} \int_{-\sqrt{4-z^2}}^{\sqrt{4-z^2}} \left(8 - 2x^2 - 2z^2\right) \mathrm{d}x\,\mathrm{d}z. \tag{16}$$

The integrand is even in x, so the inner integral is

$$8 \cdot 2h - 2 \cdot \frac{2h^3}{3} - 2z^2 \cdot 2h = \frac{8}{3}\left(4 - z^2\right)^{3/2}, \qquad h = \sqrt{4 - z^2}. \tag{17}$$

Now substitute $z = 2\sin\varphi$ with $\mathrm{d}z = 2\cos\varphi\,\mathrm{d}\varphi$. The interval $-2 \le z \le 2$ becomes $-\pi/2 \le \varphi \le \pi/2$, and $(4 - z^2)^{3/2} = (2\cos\varphi)^3 = 8\cos^3\varphi$. Then

$$V(E) = \frac{8}{3}\int_{-\pi/2}^{\pi/2} 8\cos^3\varphi \cdot 2\cos\varphi\,\mathrm{d}\varphi = \frac{128}{3}\int_{-\pi/2}^{\pi/2} \cos^4\varphi\,\mathrm{d}\varphi. \tag{18}$$

The reduction formula for the fourth power of the cosine gives

$$\int_{-\pi/2}^{\pi/2} \cos^4\varphi\,\mathrm{d}\varphi = \frac{3\pi}{8}, \qquad \text{so} \qquad V(E) = \frac{128}{3}\cdot\frac{3\pi}{8} = 16\pi. \tag{19}$$

Equation (19) agrees with Equation (15). The two calculations use different orders of integration and different antiderivatives, so they are independent checks of the limits of integration.

Numerical check. Table 1 collects four estimates. Method A is a two-dimensional Simpson rule in polar coordinates of the xz-plane. Method B is the slice method: it integrates the area of the horizontal slice at height z over $-2 \le z \le 2$. Method C is a Monte Carlo count over the box $[-2, 2] \times [0, 8] \times [-2, 2]$. Method D is a midpoint voxel sum over a grid of cells in the xz-plane.

Table 1: Numerical confirmation of the exact value 16π. The error column compares each estimate with $16\pi = 50.265482457$.

Method	Estimate	Absolute error
A: two-dimensional Simpson rule in polar coordinates	50.265482457	7.1×10^{-15}
B: slice method in z, Simpson rule	50.265488496	6.0×10^{-6}
C: Monte Carlo, $4 \cdot 10^6$ samples	50.256320000	9.2×10^{-3}
D: midpoint voxel sum, 400×400 cells	50.265506760	2.4×10^{-5}
Exact value 16π	50.265482457	—

Geometric check. The solid E lies inside the cylinder of radius 2 and height 8, whose volume is 32π. It contains the cone of radius 2 and height 8, whose volume is $32\pi/3$. Because E is bounded by a paraboloid, its volume should be half of the cylinder volume, and indeed

$$V(E) = 16\pi = \frac{1}{2}\cdot 32\pi. \tag{20}$$

The same statement holds for the average height. The base area is $\pi \cdot 2^2 = 4\pi$, so the average height of E is

$$\bar{y} = \frac{V(E)}{\mathrm{area}(D)} = \frac{16\pi}{4\pi} = 4, \tag{21}$$

which is exactly half of the peak height 8. This is the known property of a paraboloid, and it confirms the factor $2\pi \cdot 8$ of Equation (13).

The scripts `part1_step1.py` to `part1_step4.py`, `part1_figure1.py`, and `part1_figure2.py` produce every number and both pictures in this document, and each script ends with the line `PASS`.

How to solve problems of this kind

1. **Find the intersection before anything else.** Set the two bounding surfaces equal, as in Equation (1). The result is the curve that closes the solid, and it fixes the outer limits of the projection.

2. **Identify the axis of symmetry.** Both surfaces here depend on x and z only through $x^2 + z^2$, so the y-axis is an axis of symmetry and the projection is a disk. An axis of symmetry always points to polar, cylindrical, or spherical coordinates.

3. **Test simplicity along the symmetry axis.** A line parallel to the y-axis meets E in exactly one segment, so y is the inner variable. When the two surfaces are graphs over the projection, the inner integral is always the difference of the two graph functions.

4. **Check that the solid is one piece.** The difference in Equation (2) must be nonnegative on the whole projection and zero only on its boundary. A change of sign would mean that the surfaces cross inside the projection and that the solid must be split into two integrals.

5. **Write the area element of the plane.** For polar coordinates in the xz-plane the element is $r\,\mathrm{d}r\,\mathrm{d}\theta$. The extra factor r is the Jacobian of the change of variables, and it is never part of the integrand that the problem gives you.

6. **Separate the integrals when the integrand allows it.** If the integrand is a product of a function of r and a function of θ, or if one variable is missing, the double integral is a product of two single integrals. Here the integrand has no θ, so the θ integral is just 2π.

7. **Check with a second order, a number, and a geometric bound.** A different order exercises different antiderivatives. A numerical method exercises the limits. A geometric bound, such as the cylinder and the cone around E, catches large errors in the factor.